北京郵電大學

本 科 毕 业 设 计（论文）

题目：大模型驱动的代码质量多维度评估系统的设计与实现开题报告

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学　　院　　计算机学院

专　　业　　计算机科学与技术

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2026年5月

北 京 邮 电 大 学

本科毕业设计（论文）任务书

<table>
<tr><td>学院</td><td colspan="2">计算机学院（国家示范性软件学院）</td><td colspan="2">专业</td><td colspan="2">计算机科学与技术</td></tr>
<tr><td>学生姓名</td><td>宋健</td><td>学号</td><td>2022212702</td><td>班级</td><td>2022211305</td></tr>
<tr><td>指导教师姓名</td><td>金大海</td><td>所在单位</td><td>计算机学院</td><td>职称</td><td>副教授</td></tr>
<tr><td rowspan="2">设计(论文)题目</td><td colspan="6">（中文）大模型驱动的代码质量多维度评估系统的设计与实现</td></tr>
<tr><td colspan="6">（英文）Design and Implementation of Multidimensional Code Quality Evaluation System Driven by Large Models</td></tr>
<tr><td>题目类型</td><td colspan="6">工程实践类 ☑ 研究设计类☐ 理论分析类☐ 文献综述类☐ 其他☐</td></tr>
<tr><td rowspan="3">题目来源</td><td colspan="6">题目是否来源于科研项目 是☐ 否 ☑</td></tr>
<tr><td colspan="6">科研项目名称:</td></tr>
<tr><td colspan="6">科研项目负责人:</td></tr>
</table>

主要内容:

（1）分析及对比典型的大模型工具，在代码缺陷理解与分析方面选择适合的基础模型。

（2）研究确定代码质量的多维度指标，包括但不限于：功能性（是否实现需求、设计），一致性（需求、设计、编码是否一致）、正确性（是否存在代码缺陷与漏洞）、效率性（是否存在低效及冗余循环）、精简性（是否存在冗余代码）、可读性（如命名规范、注释完整性、逻辑复杂度、语句简洁性等）、可维护性（类结构是否清晰），等等。

（3）设计大模型评估流程：对每个维度设计专用提示词，并研究如何让模型生成可解释的推理过程，整合多维度结果生成综合质量报告。

（4）对比大模型评估结果与专业开发人员的人工评估，优化模型的评估一致性（如通过调整提示词减少主观偏差）。

（5）开发原型系统，构建多维度评估框架，利用大模型的跨任务能力实现“一站式”代码质量分析，并让代码质量评估结果可追溯、可理解。

主要(技术)要求:

1、对完整工程提供分析报告（不能是单一文件）；

2、支持对被测程序及大模型的管理与配置;

3、分析准确率高于 90%;

4、累计验证程序大于 20 万行代码。

主要参考文献:

[1] 陈梓豪,金大海,宫云战.基于漏洞异构图的图卷积网络漏洞检测方法[J].小型微型计算机系统,2025,46(03):697-703.DOI:10.20009/j.cnki.21-1106/TP.2023-0538.

[2] Pathak, Aditya, et al. “Rubric Is All You Need: Improving LLM-Based Code Evaluation With Question-Specific Rubrics.” Proceedings of the 2025 ACM Conference on International Computing Education Research V.1, ACM, 2025, pp. 181–95. Crossref, https://doi.org/10.1145/3702652.3744220.

[3] Wang, Ruiqi, et al. “Can LLMs Replace Human Evaluators? An Empirical Study of LLM-as-a-Judge in Software Engineering.” Proceedings of the ACM on Software Engineering, vol. 2, no. ISSTA, June 2025, pp. 1955–77. Crossref, https://doi.org/10.1145/3728963.
[4] Tong, Weixi, Tianyi Zhang, et al. “CodeJudge: Evaluating Code Generation with Large Language Models.” Accepted to EMNLP 2024 (Main, Long Paper). arXiv:2410.02184 (2024). https://doi.org/10.48550/arXiv.2410.02184.
[5] Sun, Tao, Jian Xu, et al. “BitsAI-CR: Automated Code Review via LLM in Practice.” arXiv:2501.15134 (2025). https://doi.org/10.48550/arXiv.2501.15134.
[6] Sabra, Abbas, Olivier Schmitt, Joseph Tyler, et al. “Assessing the Quality and Security of AI-Generated Code: A Quantitative Analysis.” arXiv:2508.14727 (2025). https://doi.org/10.48550/arXiv.2508.14727.
[7] Wong, Sherman, Jalaj Bhandari, et al. “A Note on Code Quality Score: LLMs for Maintainable Large Codebases.” 24 pages, ICLR format. arXiv:2508.02732 (2025). https://doi.org/10.48550/arXiv.2508.02732.

进度安排：

一、秋季学期第 11-12 周（选题与任务书阶段）

核心工作：确认选题，明确课题目标，接收并理解任务书

具体要求：

1. 本科教务管理系统完成课题选报，完成师生互选匹配；
2. 研读任务书，明确"大模型驱动代码质量多维度评估系统"核心需求，疑问及时沟通并形成书面确认；
3. 启动前期调研，梳理大模型代码分析、代码质量评估维度研究现状，收集 5 篇以上相关中英文文献并做阅读笔记。

二、秋季学期第 13-16 周（开题阶段）

核心工作：完成开题报告撰写提交，通过开题检查

具体要求：

1. 基于调研分析课题价值与研究现状，明确系统创新点；
2. 确定技术路线（大模型方案、评估框架、开发工具等），制定研究计划；
3. 按规范撰写《开题报告》并按时上传系统，提前 3 天提交初稿审核；
4. 准备答辩 PPT，清晰阐述方案，确保通过开题检查。

三、秋季学期第 17-20 周至春季学期第 1-2 周（第一阶段研究）

核心工作：完成系统需求分析与架构设计，启动大模型适配

具体要求：

1. 开展需求分析，明确系统功能与非功能需求，撰写《需求分析说明书》；
2. 设计系统整体架构，绘制架构图与模块流程图；
3. 做好质量建模，量化成果。
4. 完成大模型选型测试，对比 CodeLlama、GPT-4 Code 等模型特性，确定适配方案；
5. 完成 MVP,
5. 做对比试验，对比纯静态方法、纯大模型和“大模型驱动为核心，静态工具增强”三个方案的效果。

四、春季学期第 3-4 周（中期检查阶段）

核心工作：完成中期进展总结，通过中期检查

具体要求：

1. 按要求填写《中期进展情况检查表》，梳理阶段成果、未完成任务及后续计划；
2. 撰写《中期进展报告》，附架构图、模型测试报告等材料，提前 2 天提交审核；
3. 准备汇报内容，说明进展情况与问题解决方案，确保通过检查，计划调整需提前协商。

五、春季学期第 5-8 周（第二阶段研究）

核心工作：完成系统开发与初步测试，启动论文撰写

具体要求：

1. 基于架构实现核心模块（代码解析、大模型交互、多维度评估、结果展示）；
2. 构建测试数据集，完成系统初步测试与优化；
3. 启动论文撰写，完成摘要、引言、文献综述等章节，形成初稿框架；
4. 每两周线下沟通，演示开发进展，提交论文初稿片段审核。

六、春季学期第 9-11 周（院级审查阶段）

核心工作：完成系统优化与论文修改，通过院内查重与内审

具体要求：

1. 依据测试结果与审核意见，完成系统优化与 Bug 修复，形成可演示版本；
2. 完善论文核心章节，全文润色确保逻辑清晰；
3. 完成论文查重，确保重复率≤15%，超标及时修改；
4. 提交最终系统（含源码、说明文档）与论文初稿，配合内审并根据意见调整。

七、春季学期第 12-15 周（答辩与评优阶段）

核心工作：准备答辩材料，完成答辩及后续整改归档

具体要求：

1. 撰写答辩稿并制作 PPT，突出创新点、设计方案与成果，提前 5 天提交审核；
2. 熟悉系统演示流程，准备答辩可能涉及的问题；
3. 参加学院答辩，依据答辩意见完成论文修改；
4. 整理论文终稿、系统源码等材料归档，提交至教务系统与学院，配合成绩录入及优秀论文推荐。

注：时间节点与学校安排保持一致，若有调整另行通知。严格遵守进度计划，特殊情况无法按时完成需提前 3 个工作日提交书面申请，说明原因及调整方案。

指导教师签字		日期	

编号：_____

北京邮电大学本科生毕业设计（论文）成绩评定表

<table>
<tr><td>学生姓名</td><td colspan="2">猜猜</td><td>所在学院</td><td colspan="3">信息与通信工程学院</td></tr>
<tr><td>学号</td><td>2014210999</td><td>专业</td><td colspan="2">通信工程</td><td>班级</td><td>2014211199</td></tr>
<tr><td rowspan="2">论文题目</td><td colspan="6">（中文）猜猜看毕设题目是什么</td></tr>
<tr><td colspan="6">（英文）Just Guess What On Earth My Title Is</td></tr>
<tr><td>指导教师姓名</td><td></td><td>指导教师职称</td><td></td><td>指导教师单位</td><td colspan="2"></td></tr>
<tr><td>中期检查小组评分</td><td colspan="6">（满分 10 分）：　　中期检查小组组长签字：　　检查日期：</td></tr>
<tr><td rowspan="8">指导教师评分</td><td>评价内容</td><td colspan="2">具体要求</td><td colspan="4">分值</td><td colspan="2">评分</td></tr>
<tr><td colspan="6"></td><td>指导教师</td><td>复议</td></tr>
<tr><td>调研论证</td><td colspan="2">能独立查阅文献和从事相关调研；能正确翻译外文资料；有收集、加工各种信息及获取新知识的能力和自学能力。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td><td></td></tr>
<tr><td>方案设计</td><td colspan="2">能独立提出符合选题的可行性研究方案、实验方案、设计方案，独立进行实验（如安装、调试、操作）和研究方案论证。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td><td></td></tr>
<tr><td>能力水平</td><td colspan="2">能综合运用所学知识和技能去分析与解决毕业设计（论文）过程中遇到的实际问题；能正确处理实验数据；能对课题进行理论分析，得出有价值的结论。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td><td></td></tr>
<tr><td>学习态度</td><td colspan="2">认真、勤奋、努力、诚实、严格遵守纪律，按期饱满完成规定的任务。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td><td></td></tr>
<tr><td>设计（论文）水平</td><td colspan="2">文题相符、综述简练完整，有见解；立论正确，论述充分，结论严谨合理；实验正确，分析处理科学；文字通顺，技术用语准确，设计（论文）有理论价值和应用价值。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td><td></td></tr>
<tr><td>文本规范</td><td colspan="2">装订顺序正确，字体字号等与基本规范相符，符号统一，编号齐全，图表完备、整洁、正确。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td><td></td></tr>
<tr><td colspan="9">指导教师评分合计（满分 30 分）：
评语：

指导教师签字：　　日期：　　年　　月　　日</td></tr>
</table>

<table>
<tr><td>复议</td><td colspan="8">□是　□否　复议评分合计：　复议人签字：　复议日期：
复议有权限修改指导教师评分，选择复议后指导教师评分将由复议评分替换</td></tr>
<tr><td colspan="9">本科生毕业设计（论文）答辩成绩评定标准</td></tr>
<tr><td rowspan="8">答辩小组成绩评定</td><td>评价内容</td><td>具体要求</td><td colspan="5">分值</td><td>评分</td></tr>
<tr><td>选题</td><td>符合专业培养目标，符合社会实际、结合工程实际，难易适度，体现新颖性、综合性。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td></tr>
<tr><td>设计（论文）质量水平</td><td>全面完成任务书中规定的各项要求，文题相符，工作量饱满，写作规范，达到综合训练的要求，有理论成果和应用价值。</td><td>20</td><td>16</td><td>14</td><td>12</td><td>8</td><td></td></tr>
<tr><td>答辩准备</td><td>准备充分；有简洁、清晰、美观的演示文稿；准时到场。</td><td>5</td><td>4</td><td>3.5</td><td>3</td><td>2</td><td></td></tr>
<tr><td>内容陈述</td><td>语言表达简洁、流利、清楚、准确，思路清晰，重点突出，逻辑性强，概念清楚，论点正确；实验方法科学，分析归纳合理；结论严谨；表现出对毕业设计（论文）内容掌握透彻。</td><td>20</td><td>18</td><td>14</td><td>12</td><td>8</td><td></td></tr>
<tr><td>回答问题</td><td>回答问题准确、有深度、有理论根据、基本概念清晰。</td><td>10</td><td>8</td><td>7</td><td>6</td><td>4</td><td></td></tr>
<tr><td colspan="2">答辩小组评分合计（满分 60 分）</td><td colspan="6"></td></tr>
<tr><td colspan="8">意见：

答辩小组组长签字：_____　年　月　日</td></tr>
<tr><td colspan="8">答辩小组成员：</td></tr>
<tr><td>学院意见</td><td colspan="8">最终成绩：百分制_____；　五分制
院长签章：　学院盖章：　年　月　日</td></tr>
<tr><td>备注</td><td colspan="8"></td></tr>
</table>

注：1．毕业设计（论文）成绩由中期检查评分（满分 10 分）、指导教师评分/复议评分（满分 30 分）和答辩小组评分（满分 60 分）相加，得出百分制成绩，再按 100-90 分为“优”、89-80 分为“良”、79-70 分为“中”、69-60 分为“及格”、60 分以下为“不及格”的标准折合成五级分制成绩；

2．此表原件一式三份，一份存入学生档案，一份装订到毕业论文中，一份交教务处存入档案馆。

北 京 邮 电 大 学

本科毕业设计（论文）诚信声明

本人声明所呈交的毕业设计（论文），题目《　　》是本人在指导教师的指导下，独立进行研究工作所取得的成果。尽我所知，除了文中特别加以标注和致谢中所罗列的内容以外，论文中不包含其他人已经发表或撰写过的研究成果，也不包含为获得北京邮电大学或其他教育机构的学位或证书而使用过的材料。

申请学位论文与资料若有不实之处，本人承担一切相关责任。

本人签名：_____　　日期：

关于论文使用授权的说明

本人完全了解并同意北京邮电大学有关保留、使用学位论文的规定，即：北京邮电大学拥有以下关于学位论文的无偿使用权，具体包括：学校有权保留并向国家有关部门或机构送交学位论文，有权允许学位论文被查阅和借阅；学校可以公布学位论文的全部或部分内容，有权允许采用影印、缩印或其它复制手段保存。汇编学位论文，将学位论文的全部或部分内容编入有关数据库进行检索。（保密的学位论文在解密后遵守此规定）

本人签名：_____　　日期：

导师签名：_____　　日期：

大模型驱动的代码质量多维度评估系统的设计与实现开题报告

摘　要

这是中文摘要的部分。
它可以拥有多段。这是中文摘要的部分。
它可以拥有多段。
如果你写的太长，甚至可以到第二页。

关键词　北京邮电大学　本科生　毕业设计　模板　示例

Design and Implementation of Multidimensional Code Quality Evaluation System Driven by Large Models

ABSTRACT

This is ABSTRACT.

You can write more than one paragraph here.

If your abstract is too long, it will take up more pages.

KEY WORDS BUPT undergraduate thesis template example

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致　　谢
附　　录
外　文　资　料
外　文　译　文
开　题　报　告
中　期　检　查　表
教师指导毕业设计（论文）记录表

第一章　基础模块示例

1.1　特殊文本类型

1.1.1　脚注

社交媒体是一种供用户创建在线社群来分享信息、观点、个人信息和其它内容（如视频）的电子化交流平台，社交网络服务（social network service, SNS）和微博客（microblogging）都属于社交媒体的范畴[1]，国外较为知名的有 Facebook[1]、Instagram[2]、Twitter[3]、LinkedIn[4]等，国内较为知名的有新浪微博[5]。

在社交媒体的强覆盖下，新闻信息的传播渠道也悄然了发生变化。[2]

1.1.2　定义、定理与引理等

定义 1.1 这是一条我也不知道在说什么的定义，反正我就是写在这里做个样子罢了，也没人会仔细读。[3]

定理 1.1 这是一条我也不知道在说什么的定理，反正我就是写在这里做个样子罢了，也没人会仔细读。

公理 1.1 这是一条我也不知道在说什么的公理，反正我就是写在这里做个样子罢了，也没人会仔细读。

引理 1.1 这是一条我也不知道在说什么的引理，反正我就是写在这里做个样子罢了，也没人会仔细读。

命题 1.1 这是一条我也不知道在说什么的命题，反正我就是写在这里做个样子罢了，也没人会仔细读。

推论 1.1 这是一条我也不知道在说什么的推论，反正我就是写在这里做个样子罢了，也没人会仔细读。

1.1.3　中英文文献、学位论文引用

根据美国皮尤研究中心的 2017 年 9 月发布的调查结果[4]，67% 的美国民众会从社交媒体上获取新闻信息，其中高使用频率用户占 20%。在国内，中国互联网信息中心

本项目来源于科研项目“基于 LaTeX 的本科毕业设计”，项目编号 1124

[1] http://www.facebook.com/

[2] https://www.instagram.com/

[3] http://www.twitter.com/

[4] http://www.linkedin.com/

[5] http://www.weibo.com/

《2016 年中国互联网新闻市场研究报告》[5] 也显示，社交媒体已逐渐成为新闻获取、评论、转发、跳转的重要渠道，在 2016 年下半年，曾经通过社交媒体获取过新闻资讯的用户比例高达 90.7%，在微信、微博等社交媒体参与新闻评论的比例分别为 62.8% 和 50.2%。社交媒体正在成为网络上热门事件生成并发酵的源头，在形成传播影响力后带动传统媒体跟进报道，最终形成更大规模的舆论浪潮。[6]

在国内，新浪微博由于其发布方便、传播迅速、受众广泛且总量大的特点，成为了虚假信息传播的重灾区：《中国新媒体发展报告（2013）》[7] 显示，2012 年的 100 件微博热点舆情案例中，有超过 1/3 出现谣言；《中国新媒体发展报告（2015）》[8] 对 2014 年传播较广、比较典型的 92 条假新闻进行了多维度分析，发现有 59% 的虚假新闻首发于新浪微博。

此等信息的传播严重损害了有关公众人物的名誉权，降低了社交媒体服务商的商业美誉度，扰乱了网络空间秩序，冲击着网民的认知，极易对民众造成误导，带来诸多麻烦和经济损失，甚至会导致社会秩序的混乱。针对社交媒体谣言采取行动成为了有关部门、服务提供商和广大民众的共同选择。[3]

1.2 图表及其引用

此处引用了简单的表 1-1 。

请注意，LaTeX 的图表排版规则决定了图表**不一定会乖乖呆在你插入的地方**，这是为了避免 Word 中由于图片尺寸不匹配在页面下部出现的的空白，所以请不要使用“下图”“下表”作为指向文字，应使用“图 1-1 所示”这样的表述。

表 1-1 基于浏览者行为的特征

特征	**描述**	**形式与理论范围**
点赞量	微博的点赞数量	数值，$\mathbb{N}$
评论量	微博的评论数量	数值，$\mathbb{N}$
转发量	微博的转发数量	数值，$\mathbb{N}$

此处引用了复杂的表 1-2 。

表 1-2 基于浏览者行为的复杂特征

类别	**特征**	**不知道叫什么的表头**	
		描述	**形式与理论范围**
正常互动	点赞量	微博的点赞数量	数值，$\mathbb{N}$
	评论量	微博的评论数量	数值，$\mathbb{N}$
	转发量	微博的转发数量	数值，$\mathbb{N}$
非正常互动	羡慕量	微博的羡慕数量	数值，$\mathbb{N}$

此处展示了更专业的表 1-3 ，一个好的表格没有竖线。

表 1-3 红警 2 名词解释

术语类别	缩略语	解释
游戏	兵营	兵营（Barracks），《命令与征服 红色警戒 2：尤里的复仇》游戏中的一种生产建筑，用以生产步兵单位
	建造场	建造场（Construction Yard），《命令与征服 红色警戒 2：尤里的复仇》游戏中的一种基础建筑，用以支持其他建筑的建造
	矿厂	矿石精炼厂（Ore Refinery），《命令与征服 红色警戒 2：尤里的复仇》游戏中的一种资源建筑，用以将矿车采集的矿石转化为游戏资金
	空指	空指部（Airforce Command Headquarters），《命令与征服 红色警戒 2：尤里的复仇》游戏中的一种资源建筑，用以提供雷达功能和 T2 科技及生产部分空军单位
	相机	游戏术语，特指游戏内的观察区域和视角
	重工	战车工厂（War Factory），《命令与征服 红色警戒 2：尤里的复仇》游戏中的一种生产建筑，用以生产载具单位
	战争迷雾	游戏术语，《命令与征服 红色警戒 2：尤里的复仇》中指黑色的未探索区域

此处引用了一张图。图 1-1 表示的是一个由含有 4 个神经元的输入层、含有 3 个神经元的隐藏层和含有 4 个神经元的输出层组成的自编码器，+1 代表偏置项。

1.3 公式与算法表示

1.3.1 例子：基于主成分分析

1.3.1.1 主成分分析算法

下面对主成分分析进行介绍。

主成分分析是一种简单的机器学习算法，其功能可以从两方面解释：一方面可以认为它提供了一种压缩数据的方式，另一方面也可以认为它是一种学习数据表示的无监督学习算法。[9] 通过 PCA，我们可以得到一个恰当的超平面及一个投影矩阵，通过投影矩阵，样本点将被投影在这一超平面上，且满足最大可分性（投影后样本点的方差最大化），直观上讲，也就是能尽可能分开。

对中心化后的样本点集 $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_m\}$（有 $\sum_{i=1}^{m} \boldsymbol{x}_i = 0$），考虑将其最大可分地投影到新坐标系 $\boldsymbol{W} = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_i, \ldots, \boldsymbol{w}_d\}$，其中 $\boldsymbol{w}_i$ 是标准正交基向量，满足 $\|\boldsymbol{w}_i\|_2 = 1$，$\boldsymbol{w}_i^T \boldsymbol{w}_j = 0$（$i \neq j$）。假设我们需要 d'（$d' < d$）个主成分，那么样

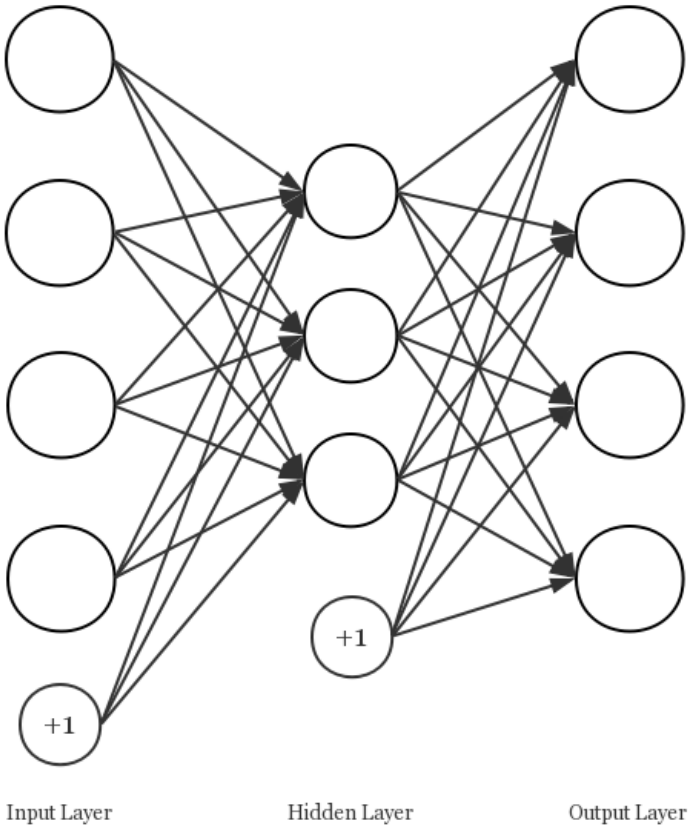

图 1-1 自编码器结构

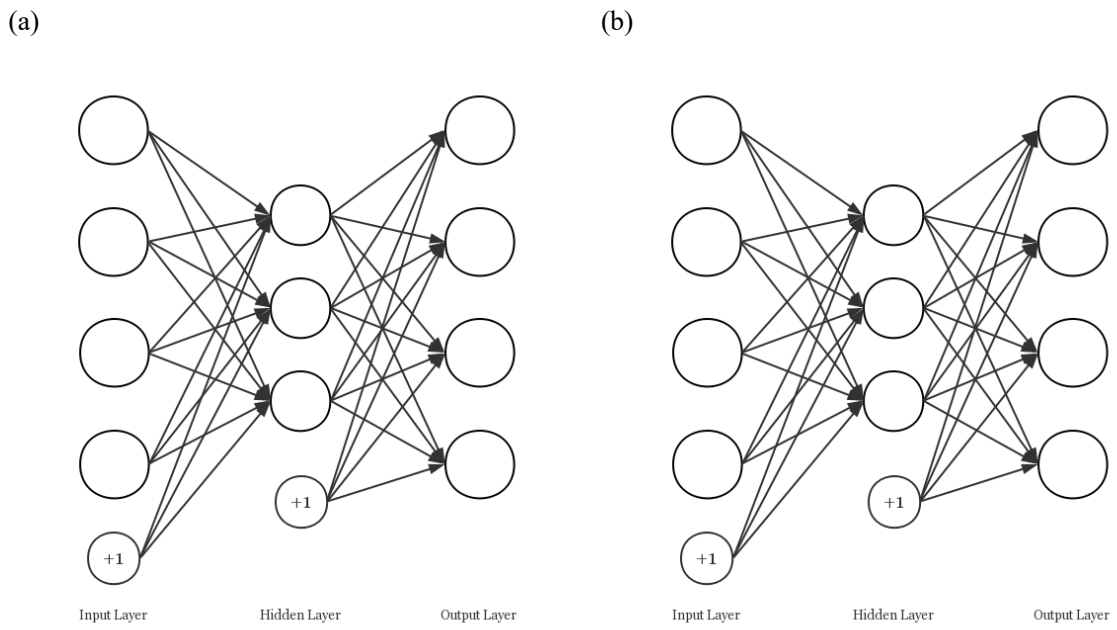

图 1-2 这是两个自编码器结构，我就是排一下子图的效果：(a)左边的自编码器，(b)右边的自编码器

本点 $\boldsymbol{x}_i$ 在低维坐标系中的投影是 $\boldsymbol{z}_i = (z_{i1}; z_{i2}; \ldots; z_{id'})$，其中 $z_{ij} = \boldsymbol{w}_j^{\mathrm{T}} \boldsymbol{x}_i$，是 $\boldsymbol{x}_i$ 在低维坐标系下第 j 维的坐标。对整个样本集，投影后样本点的方差是

$$\begin{aligned}
&\frac{1}{m}\sum_{i=1}^{m} \boldsymbol{z}_i^{\mathrm{T}} \boldsymbol{z}_i \\
=&\frac{1}{m}\sum_{i=1}^{m} (\boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{W})^{\mathrm{T}} (\boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{W}) \\
=&\frac{1}{m}\sum_{i=1}^{m} \boldsymbol{W}^{\mathrm{T}} \boldsymbol{x}_i \boldsymbol{x}_i^{\mathrm{T}} \boldsymbol{W} \\
=&\frac{1}{m} \boldsymbol{W}^{\mathrm{T}} \boldsymbol{X} \boldsymbol{X}^{\mathrm{T}} \boldsymbol{W}
\end{aligned} \tag{1-1}$$

由于我们知道新坐标系 $\boldsymbol{W}$ 的列向量是标准正交基向量，且样本点集 $\boldsymbol{X}$ 已经过中心化，则 PCA 的优化目标可以写为

$$\begin{aligned}
\max_{\boldsymbol{W}} \quad & tr(\boldsymbol{W}^{\mathrm{T}} \boldsymbol{X} \boldsymbol{X}^{\mathrm{T}} \boldsymbol{W}) \\
\text{s.t.} \quad & \boldsymbol{W}^{\mathrm{T}} \boldsymbol{W} = \boldsymbol{I}
\end{aligned} \tag{1-2}$$

由于 $\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}$ 是协方差矩阵，那么只需对它做特征值分解，即

$$\boldsymbol{X}^{\mathrm{T}} \boldsymbol{X} = \boldsymbol{W} \boldsymbol{\Lambda} \boldsymbol{W}^{\mathrm{T}} \tag{1-3}$$

其中 $\boldsymbol{\Lambda} = diag(\boldsymbol{\lambda})$，$\boldsymbol{\lambda} = \{\lambda_1, \lambda_2, \ldots, \lambda_m\}$。

具体地，考虑到它是半正定矩阵的二次型，存在最大值，可对式（1-2 ）使用拉格朗日乘数法

$$\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}} \boldsymbol{w}_i = \lambda_i \boldsymbol{w}_i \tag{1-4}$$

之后将求得的特征值降序排列，取前 d' 个特征值对应的特征向量组成所需的投影矩阵 $\boldsymbol{W}' = (\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_{d'})$，即可得到 PCA 的解。PCA 算法的描述如算法1所示。

算法 1 主成分分析（PCA）

输入： 样本集 $\boldsymbol{x} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_i, \ldots, \boldsymbol{x}_m\}$，低维空间维数 d'

输出： 投影矩阵 $\boldsymbol{W}' = (\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_{d'})$

1: 对所有样本中心化 $\boldsymbol{x}_i \leftarrow \boldsymbol{x}_i - \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{x}_i$

2: 计算样本的协方差 $\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}$

3: 对协方差矩阵 $\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}$ 做特征值分解

4: 取最大的 d' 个特征值所对应的特征向量 $\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_{d'}$

1.3.1.2 主成分分析可信度评估方法

记待判定微博 $\boldsymbol{w}_0$ 的经典特征向量为 $\boldsymbol{f}_0^c$，它的发布者在 $\boldsymbol{w_0}$ 前发布的 k 条微博为 $\boldsymbol{W} = \boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_k$，这 k 条微博对应的经典特征向量集为 $\boldsymbol{F}_W^c = \{\boldsymbol{f}_1^c, \boldsymbol{f}_2^c, \ldots, \boldsymbol{f}_k^c\}$。令 $label = 1$ 代表谣言，$label = 0$ 代表非谣言。算法的具体流程如算法2所示。

算法 2 基于 PCA 的信息可信度评估

输入： $\boldsymbol{f}_0^c$，$\boldsymbol{F}_W^c$，保留主成分数 n

输出： 标签 $label \in \{0, 1\}$

对所有特征向量应用 PCA，保留前 n 个主成分 $\boldsymbol{o}_i^c \leftarrow PCA(\boldsymbol{f}_i^c, n)$ $(i = 0, 1, \ldots, k)$
计算 $\boldsymbol{F}_W^c$ 中各向量的平均距离 μ 和标准差 σ
计算阈值 $thr = \mu/\sigma$
if $\min_{1<j\leq k} \|\boldsymbol{o}_0^c - \boldsymbol{o}_j^c\|_2 > thr$ **then**
    $label \leftarrow 1$
else
    $label \leftarrow 0$
end if

1.4 代码表示

1.4.1 直接书写代码在.tex 中

下面的代码是用 Python 编写的加法函数。

代码 加法

```python
def plusFunc(a, b):
  return a + b
```

1.4.2 引用代码文件

下面的代码是用 Python 文件中引入的倒序打印 x 到 1 的函数，请查看 code 文件夹。

代码 倒序打印数字

```python
def numbers(x):
    if x > 0:
        print(x)
        numbers(x-1)
```

1.5 列表样式

1.5.1 使用圆点作为项目符号

- **第一章为基础模块示例**，是的，本章的名字就是基础模块示例，正如你看到这个样子。
- **第二章为不存在**，是的，其实它不存在。

1.5.2 使用数字作为项目符号

1. **第一章为基础模块示例**，是的，本章的名字就是基础模块示例，正如你看到这个样子。
2. **第二章为不存在**，是的，其实它不存在。

1.5.3 句中数字编号列表样式

1. **第一章为基础模块示例**，是的，本章的名字就是基础模块示例，正如你看到这个样子;2. **第二章为不存在**，是的，其实它不存在。

第二章　为了目录撑到第二页

2.1　我不得不再添加一点内容

2.2　尽管这些章节一点正文都没有

2.3　是的

2.4　真的没有

2.5　我已经不知道说什么了

2.6　如果有，我们就祝愿一下学校教务处什么时候转变一下思维

2.7　把控制格式这种事情往前做

2.8　不要总是觉得折磨学生是合理的

2.9　你拿着教学管理岗位的工资

2.10　你需要折磨一下你自己才对

2.11　不要觉得我对别人要求太高，对自己太低

2.12　我对自己要求低的话也不至于想要修订这份模板

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致　　谢

此处请写致谢的内容。
它可以有多段。

附　　录

附录 1　缩略语表

表 附-1　基于浏览者行为的特征

特征	描述	形式与理论范围
点赞量	微博的点赞数量	数值，$\mathbb{N}$
评论量	微博的评论数量	数值，$\mathbb{N}$
转发量	微博的转发数量	数值，$\mathbb{N}$

表 附-2　基于浏览者行为的复杂特征

类别	特征	不知道叫什么的表头	
		描述	形式与理论范围
正常互动	点赞量	微博的点赞数量	数值，$\mathbb{N}$
	评论量	微博的评论数量	数值，$\mathbb{N}$
	转发量	微博的转发数量	数值，$\mathbb{N}$
非正常互动	羡慕量	微博的羡慕数量	数值，$\mathbb{N}$

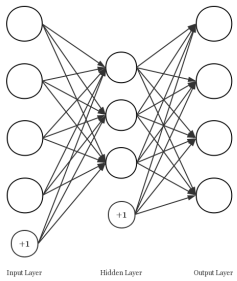

图 附-1　自编码器结构

代码 附-1 减法

```
def minusFunc(a, b):
  return a - b
```

$$\max_{\boldsymbol{W}} tr(\boldsymbol{W}^{\mathrm{T}}\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}\boldsymbol{W}) \qquad \text{式（附-1）}$$

附录 2　数学符号

数和数组

a	标量（整数或实数）
$\boldsymbol{a}$	向量
$dim()$	向量的维数
$\boldsymbol{A}$	矩阵
$\boldsymbol{A}^{\mathrm{T}}$	矩阵 **A** 的转置
$\boldsymbol{I}$	单位矩阵（维度依据上下文而定）
$diag(\boldsymbol{a})$	对角方阵，其中对角元素由向量 $\boldsymbol{a}$ 确定

外 文 资 料

SOCIAL SCIENCE

The spread of true and false news online

Soroush Vosoughi,[1] Deb Roy,[1] Sinan Aral[2]*

We investigated the differential diffusion of all of the verified true and false news stories distributed on Twitter from 2006 to 2017. The data comprise ~126,000 stories tweeted by ~3 million people more than 4.5 million times. We classified news as true or false using information from six independent fact-checking organizations that exhibited 95 to 98% agreement on the classifications. Falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information, and the effects were more pronounced for false political news than for false news about terrorism, natural disasters, science, urban legends, or financial information. We found that false news was more novel than true news, which suggests that people were more likely to share novel information. Whereas false stories inspired fear, disgust, and surprise in replies, true stories inspired anticipation, sadness, joy, and trust. Contrary to conventional wisdom, robots accelerated the spread of true and false news at the same rate, implying that false news spreads more than the truth because humans, not robots, are more likely to spread it.

Foundational theories of decision-making (*1*–*3*), cooperation (*4*), communication (*5*), and markets (*6*) all view some conceptualization of truth or accuracy as central to the functioning of nearly every human endeavor. Yet, both true and false information spreads rapidly through online media. Defining what is true and false has become a common political strategy, replacing debates based on a mutually agreed on set of facts. Our economies are not immune to the spread of falsity either. False rumors have affected stock prices and the motivation for large-scale investments, for example, wiping out $130 billion in stock value after a false tweet claimed that Barack Obama was injured in an explosion (*7*). Indeed, our responses to everything from natural disasters (*8*, *9*) to terrorist attacks (*10*) have been disrupted by the spread of false news online.

New social technologies, which facilitate rapid information sharing and large-scale information cascades, can enable the spread of misinformation (i.e., information that is inaccurate or misleading). But although more and more of our access to information and news is guided by these new technologies (*11*), we know little about their contribution to the spread of falsity online. Though considerable attention has been paid to anecdotal analyses of the spread of false news by the media (*12*), there are few large-scale empirical investigations of the diffusion of misinformation or its social origins. Studies of the spread of misinformation are currently limited to analyses of small, ad hoc samples that ignore two of the most important scientific questions: How do truth and falsity diffuse differently, and what factors of human judgment explain these differences?

Current work analyzes the spread of single rumors, like the discovery of the Higgs boson (*13*) or the Haitian earthquake of 2010 (*14*), and multiple rumors from a single disaster event, like the Boston Marathon bombing of 2013 (*10*), or it develops theoretical models of rumor diffusion (*15*), methods for rumor detection (*16*), credibility evaluation (*17*, *18*), or interventions to curtail the spread of rumors (*19*). But almost no studies comprehensively evaluate differences in the spread of truth and falsity across topics or examine why false news may spread differently than the truth. For example, although Del Vicario *et al.* (*20*) and Bessi *et al.* (*21*) studied the spread of scientific and conspiracy-theory stories, they did not evaluate their veracity. Scientific and conspiracy-theory stories can both be either true or false, and they differ on stylistic dimensions that are important to their spread but orthogonal to their veracity. To understand the spread of false news, it is necessary to examine diffusion after differentiating true and false scientific stories and true and false conspiracy-theory stories and controlling for the topical and stylistic differences between the categories themselves. The only study to date that segments rumors by veracity is that of Friggeri *et al.* (*19*), who analyzed ~4000 rumors spreading on Facebook and focused more on how fact checking affects rumor propagation than on how falsity diffuses differently than the truth (*22*).

In our current political climate and in the academic literature, a fluid terminology has arisen around "fake news," foreign interventions in U.S. politics through social media, and our understanding of what constitutes news, fake news, false news, rumors, rumor cascades, and other related terms. Although, at one time, it may have been appropriate to think of fake news as referring to the veracity of a news story, we now believe that this phrase has been irredeemably polarized in our current political and media climate. As politicians have implemented a political strategy of labeling news sources that do not support their positions as unreliable or fake news, whereas sources that support their positions are labeled reliable or not fake, the term has lost all connection to the actual veracity of the information presented, rendering it meaningless for use in academic classification. We have therefore explicitly avoided the term fake news throughout this paper and instead use the more objectively verifiable terms "true" or "false" news. Although the terms fake news and misinformation also imply a willful distortion of the truth, we do not make any claims about the intent of the purveyors of the information in our analyses. We instead focus our attention on veracity and stories that have been verified as true or false.

We also purposefully adopt a broad definition of the term news. Rather than defining what constitutes news on the basis of the institutional source of the assertions in a story, we refer to any asserted claim made on Twitter as news (we defend this decision in the supplementary materials section on "reliable sources," section S1.2). We define news as any story or claim with an assertion in it and a rumor as the social phenomena of a news story or claim spreading or diffusing through the Twitter network. That is, rumors are inherently social and involve the sharing of claims between people. News, on the other hand, is an assertion with claims, whether it is shared or not.

A rumor cascade begins on Twitter when a user makes an assertion about a topic in a tweet, which could include written text, photos, or links to articles online. Others then propagate the rumor by retweeting it. A rumor's diffusion process can be characterized as having one or more cascades, which we define as instances of a rumor-spreading pattern that exhibit an unbroken retweet chain with a common, singular origin. For example, an individual could start a rumor cascade by tweeting a story or claim with an assertion in it, and another individual could independently start a second cascade of the same rumor (pertaining to the same story or claim) that is completely independent of the first cascade, except that it pertains to the same story or claim. If they remain independent, they represent two cascades of the same rumor. Cascades can be as small as size one (meaning no one retweeted the original tweet). The number of cascades that make up a rumor is equal to the number of times the story or claim was independently tweeted by a user (not retweeted). So, if a rumor "A" is tweeted by 10 people separately, but not retweeted, it would have 10 cascades, each of size one. Conversely, if a second rumor "B" is independently tweeted by two people and each of those two tweets is retweeted 100 times, the rumor would consist of two cascades, each of size 100.

Here we investigate the differential diffusion of true, false, and mixed (partially true, partially false) news stories using a comprehensive data set of all of the fact-checked rumor cascades that spread on Twitter from its inception in 2006 to 2017. The data include ~126,000 rumor cascades spread by ~3 million people more than 4.5 million times. We sampled all rumor cascades investigated by six independent fact-checking organizations

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(snopes.com, politifact.com, factcheck.org, truthorfiction.com, hoax-slayer.com, and urbanlegends.about.com) by parsing the title, body, and verdict (true, false, or mixed) of each rumor investigation reported on their websites and automatically collecting the cascades corresponding to those rumors on Twitter. The result was a sample of rumor cascades whose veracity had been agreed on by these organizations between 95 and 98% of the time. We cataloged the diffusion of the rumor cascades by collecting all English-language replies to tweets that contained a link to any of the aforementioned websites from 2006 to 2017 and used optical character recognition to extract text from images where needed. For each reply tweet, we extracted the original tweet being replied to and all the retweets of the original tweet. Each retweet cascade represents a rumor propagating on Twitter that has been verified as true or false by the fact-checking organizations (see the supplementary materials for more details on cascade construction). We then quantified the cascades' depth (the number of retweet hops from the origin tweet over time, where a hop is a retweet by a new unique user), size (the number of users involved in the cascade over time), maximum breadth (the maximum number of users involved in the cascade at any depth), and structural virality (*23*) (a measure that interpolates between content spread through a single, large broadcast and that which spreads through multiple generations, with any one individual directly responsible for only a fraction of the total spread) (see the supplementary materials for more detail on the measurement of rumor diffusion).

As a rumor is retweeted, the depth, size, maximum breadth, and structural virality of the cascade increase (Fig. 1A). A greater fraction of false rumors experienced between 1 and 1000 cascades, whereas a greater fraction of true rumors experienced more than 1000 cascades (Fig. 1B); this was also true for rumors based on political news (Fig. 1D). The total number of false rumors peaked at the end of both 2013 and 2015 and again at the end of 2016, corresponding to the last U.S. presidential election (Fig. 1C). The data also show clear increases in the total number of false political rumors during the 2012 and 2016 U.S. presidential elections (Fig. 1E) and a spike in rumors that contained partially true and partially false information during the Russian annexation of Crimea in 2014 (Fig. 1E). Politics was the largest rumor category in our data, with ~45,000 cascades, followed by urban legends, business, terrorism, science, entertainment, and natural disasters (Fig. 1F).

When we analyzed the diffusion dynamics of true and false rumors, we found that falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information [Kolmogorov-Smirnov (K-S) tests are reported in tables S3 to S10]. A significantly greater fraction of false cascades than true cascades exceeded a depth of 10, and the top 0.01% of false cascades diffused eight hops deeper into the Twittersphere than the truth, diffusing to depths

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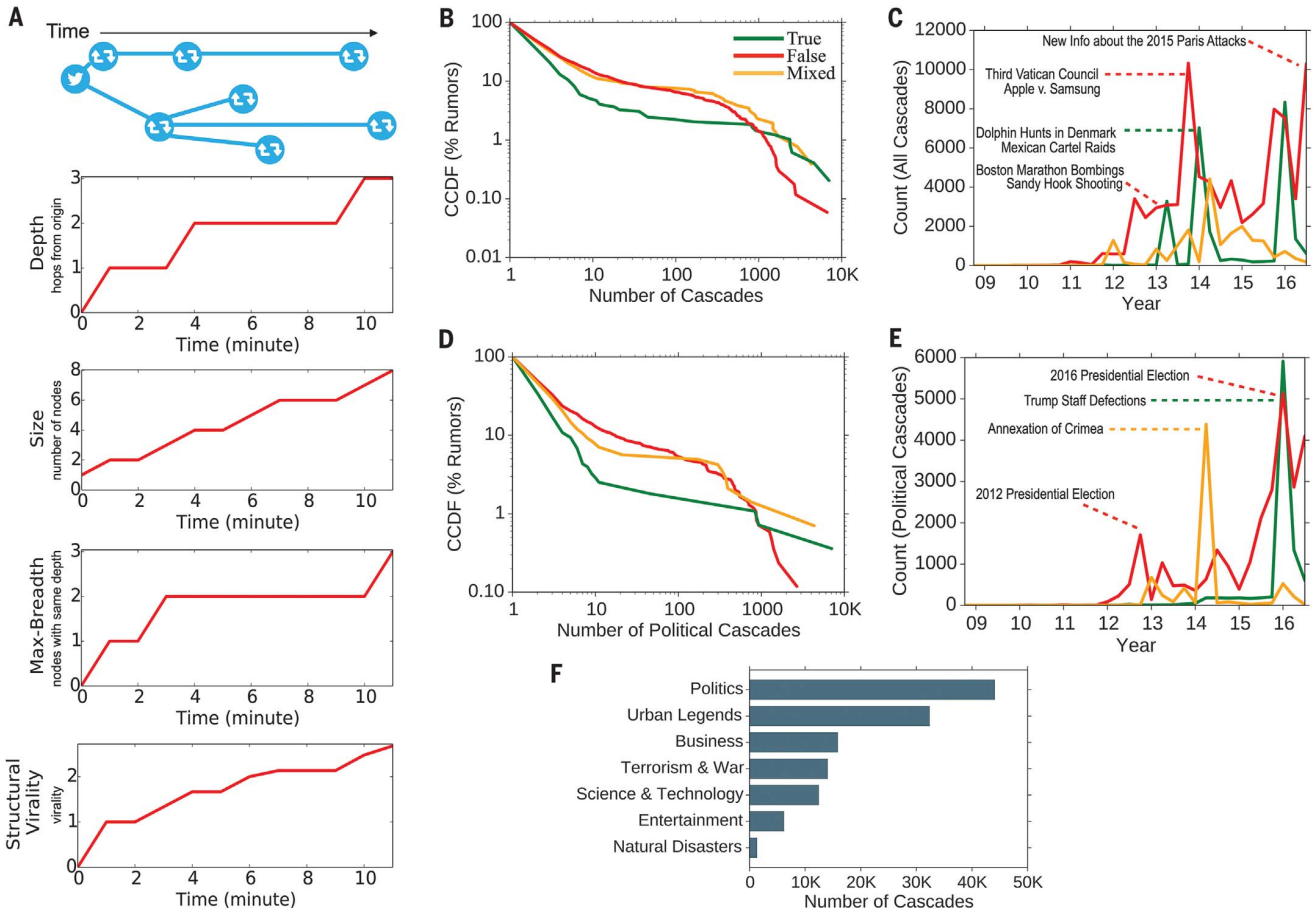

Fig. 1. Rumor cascades. (**A**) An example rumor cascade collected by our method as well as its depth, size, maximum breadth, and structural virality over time. "Nodes" are users. (**B**) The complementary cumulative distribution functions (CCDFs) of true, false, and mixed (partially true and partially false) cascades, measuring the fraction of rumors that exhibit a given number of cascades. (**C**) Quarterly counts of all true, false, and mixed rumor cascades that diffused on Twitter between 2006 and 2017, annotated with example rumors in each category. (**D**) The CCDFs of true, false, and mixed political cascades. (**E**) Quarterly counts of all true, false, and mixed political rumor cascades that diffused on Twitter between 2006 and 2017, annotated with example rumors in each category. (**F**) A histogram of the total number of rumor cascades in our data across the seven most frequent topical categories.

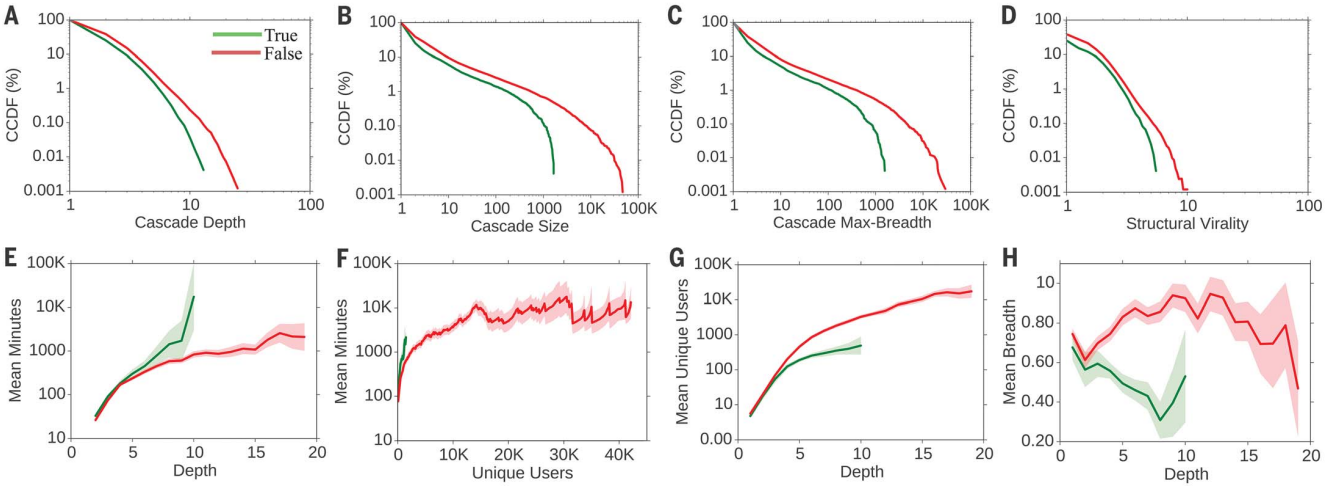

Fig. 2. Complementary cumulative distribution functions (CCDFs) of true and false rumor cascades. (**A**) Depth. (**B**) Size. (**C**) Maximum breadth. (**D**) Structural virality. (**E** and **F**) The number of minutes it takes for true and false rumor cascades to reach any (E) depth and (F) number of unique Twitter users. (**G**) The number of unique Twitter users reached at every depth and (**H**) the mean breadth of true and false rumor cascades at every depth. In (H), plot is lognormal. Standard errors were clustered at the rumor level (i.e., cascades belonging to the same rumor were clustered together; see supplementary materials for additional details).

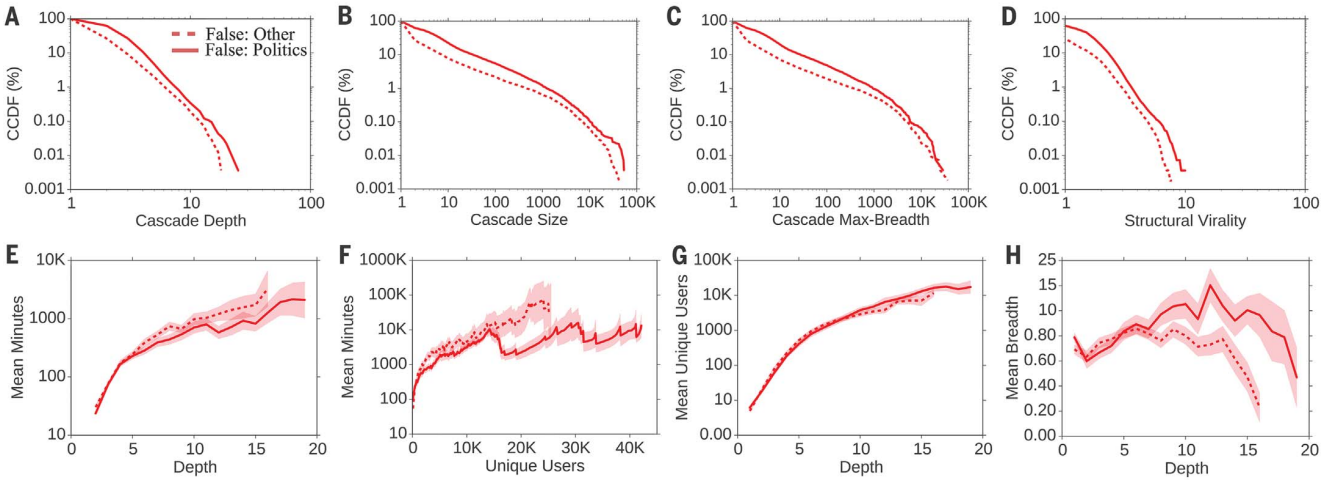

Fig. 3. Complementary cumulative distribution functions (CCDFs) of false political and other types of rumor cascades. (**A**) Depth. (**B**) Size. (**C**) Maximum breadth. (**D**) Structural virality. (**E** and **F**) The number of minutes it takes for false political and other false news cascades to reach any (E) depth and (F) number of unique Twitter users. (**G**) The number of unique Twitter users reached at every depth and (**H**) the mean breadth of these false rumor cascades at every depth. In (H), plot is lognormal. Standard errors were clustered at the rumor level.

greater than 19 hops from the origin tweet (Fig. 2A). Falsehood also reached far more people than the truth. Whereas the truth rarely diffused to more than 1000 people, the top 1% of false-news cascades routinely diffused to between 1000 and 100,000 people (Fig. 2B). Falsehood reached more people at every depth of a cascade than the truth, meaning that many more people retweeted falsehood than they did the truth (Fig. 2C). The spread of falsehood was aided by its virality, meaning that falsehood did not simply spread through broadcast dynamics but rather through peer-to-peer diffusion characterized by a viral branching process (Fig. 2D).

It took the truth about six times as long as falsehood to reach 1500 people (Fig. 2F) and 20 times as long as falsehood to reach a cascade depth of 10 (Fig. 2E). As the truth never diffused beyond a depth of 10, we saw that falsehood reached a depth of 19 nearly 10 times faster than the truth reached a depth of 10 (Fig. 2E). Falsehood also diffused significantly more broadly (Fig. 2H) and was retweeted by more unique users than the truth at every cascade depth (Fig. 2G).

False political news (Fig. 1D) traveled deeper (Fig. 3A) and more broadly (Fig. 3C), reached more people (Fig. 3B), and was more viral than any other category of false information (Fig. 3D). False political news also diffused deeper more quickly (Fig. 3E) and reached more than 20,000 people nearly three times faster than all other types of false news reached 10,000 people (Fig. 3F). Although the other categories of false news reached about the same number of unique users at depths between 1 and 10, false political news routinely reached the most unique users at depths greater than 10 (Fig. 3G). Although all other categories of false news traveled slightly more broadly at shallower depths, false political news traveled more broadly at greater depths, indicating that more-popular false political news items exhibited broader and more-accelerated diffusion dynamics

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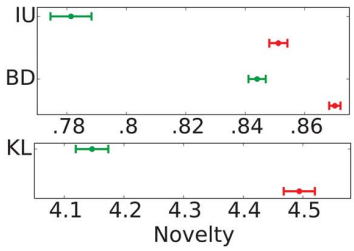
A

	median		mean		mean (log)		stdv (log)		ks-test
	false	true	false	true	false	true	false	true	
followers	410	466	2234	5240	2.62	2.68	0.69	0.88	D=0.104, p∼0.0
followees	383	509	1002	1707	2.59	2.72	0.85	0.96	D=0.136, p∼0.0
verified	0	0	0.002	0.006	nd	nd	nd	nd	D=0.005, p<0.001
engagement	9.52	9.54	19.70	24.65	0.91	0.90	0.65	0.76	D=0.054, p∼0.0
account age	982	1214	1072	1269	2.90	2.97	0.39	0.42	D=0.125, p∼0.0

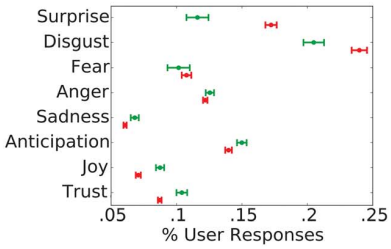
B

	coef	odds ratio	std err	z	P>|z|	[95.0% Conf. Int.]	
account age	0.0002	1.000160	2.07e-05	7.759	0.000	0.000	0.000
engagement	0.0066	1.006648	0.000	18.019	0.000	0.006	0.007
falsehood	0.5350	1.707489	0.084	6.366	0.000	0.370	0.700
followees	-1.639e-05	0.999984	8.73e-06	-1.877	0.060	-3.35e-05	7.22e-07
followers	5.192e-05	1.000052	7.77e-06	6.682	0.000	3.67e-05	6.72e-05
intercept	-2.3941	0.091257	0.072	-33.437	0.000	-2.534	-2.254
verified	1.4261	4.162467	0.090	15.915	0.000	1.250	1.602

C



D



E

	mean		variance		ks-test
	false	true	false	true	
IU	0.85	0.78	0.0052	0.0072	D=0.457, p∼0.0
KL	4.49	4.15	0.1618	0.0948	D=0.433, p∼0.0
BD	0.87	0.84	0.0008	0.0008	D=0.415, p∼0.0

F

	mean		variance		ks-test
	false	true	false	true	
surprise	0.172	0.116	0.0167	0.0072	D=0.205, p∼0.0
disgust	0.240	0.205	0.0260	0.0227	D=0.102, p∼0.0
fear	0.108	0.102	0.0120	0.0095	D=0.021, p∼0.164
anger	0.122	0.126	0.0074	0.0111	D=0.023, p∼0.078
sadness	0.061	0.068	0.0038	0.0065	D=0.037, p∼0.0
anticipation	0.140	0.150	0.0093	0.0154	D=0.038, p∼0.0
joy	0.071	0.087	0.0054	0.0104	D=0.061, p∼0.0
trust	0.087	0.104	0.0058	0.0119	D=0.060, p∼0.0

Fig. 4. Models estimating correlates of news diffusion, the novelty of true and false news, and the emotional content of replies to news. (**A**) Descriptive statistics on users who participated in true and false rumor cascades as well as K-S tests of the differences in the distributions of these measures across true and false rumor cascades. (**B**) Results of a logistic regression model estimating users' likelihood of retweeting a rumor as a function of variables shown at the left. coeff, logit coefficient; *z*, *z* score. (**C**) Differences in the information uniqueness (IU), scaled Bhattacharyya distance (BD), and K-L divergence (KL) of true (green) and false (red) rumor tweets compared to the corpus of prior tweets the user was exposed to in the 60 days before retweeting the rumor tweet. (**D**) The emotional content of replies to true (green) and false (red) rumor tweets across seven dimensions categorized by the NRC. (**E**) Mean and variance of the IU, KL, and BD of true and false rumor tweets compared to the corpus of prior tweets the user has seen in the 60 days before seeing the rumor tweet as well as K-S tests of their differences across true and false rumors. (**F**) Mean and variance of the emotional content of replies to true and false rumor tweets across seven dimensions categorized by the NRC as well as K-S tests of their differences across true and false rumors. All standard errors are clustered at the rumor level, and all models are estimated with cluster-robust standard errors at the rumor level.

(Fig. 3H). Analysis of all news categories showed that news about politics, urban legends, and science spread to the most people, whereas news about politics and urban legends spread the fastest and were the most viral in terms of their structural virality (see fig. S11 for detailed comparisons across all topics).

One might suspect that structural elements of the network or individual characteristics of the users involved in the cascades explain why falsity travels with greater velocity than the truth. Perhaps those who spread falsity "followed" more people, had more followers, tweeted more often, were more often "verified" users, or had been on Twitter longer. But when we compared users involved in true and false rumor cascades, we found that the opposite was true in every case. Users who spread false news had significantly fewer followers (K-S test = 0.104, $P \sim 0.0$), followed significantly fewer people (K-S test = 0.136, $P \sim 0.0$), were significantly less active on Twitter (K-S test = 0.054, $P \sim 0.0$), were verified significantly less often (K-S test = 0.004, $P < 0.001$), and had been on Twitter for significantly less time (K-S test = 0.125, $P \sim 0.0$) (Fig. 4A). Falsehood diffused farther and faster than the truth despite these differences, not because of them.

When we estimated a model of the likelihood of retweeting, we found that falsehoods were 70% more likely to be retweeted than the truth (Wald chi-square test, $P \sim 0.0$), even when controlling for the account age, activity level, and number of followers and followees of the original tweeter, as well as whether the original tweeter was a verified user (Fig. 4B). Because user characteristics and network structure could not explain the differential diffusion of truth and falsity, we sought alternative explanations for the differences in their diffusion dynamics.

One alternative explanation emerges from information theory and Bayesian decision theory. Novelty attracts human attention (*24*), contributes to productive decision-making (*25*), and encourages information sharing (*26*) because novelty updates our understanding of the world. When information is novel, it is not only surprising, but also more valuable, both from an information theoretic perspective [in that it provides the greatest aid to decision-making (*25*)] and from a social perspective [in that it conveys social status on one that is "in the know" or has access to unique "inside" information (*26*)]. We therefore tested whether falsity was more novel than the truth and whether Twitter users were more likely to retweet information that was more novel.

To assess novelty, we randomly selected ~5000 users who propagated true and false rumors and extracted a random sample of ~25,000 tweets that they were exposed to in the 60 days prior to their decision to retweet a rumor. We then specified a latent Dirichlet Allocation Topic model (*27*), with 200 topics and trained on 10 million English-language tweets, to calculate the information distance between the rumor tweets and all the prior tweets that users were exposed to before retweeting the rumor tweets. This generated a probability distribution over the 200 topics for each tweet in our data set. We then measured how novel the information in the true and false rumors was by comparing the topic distributions of the rumor tweets with the topic distributions of the tweets to which users were exposed in the 60 days before their retweet. We found that false rumors were significantly more

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novel than the truth across all novelty metrics, displaying significantly higher information uniqueness (K-S test = 0.457, $P \sim 0.0$) (*28*), Kullback-Leibler (K-L) divergence (K-S test = 0.433, $P \sim 0.0$) (*29*), and Bhattacharyya distance (K-S test = 0.415, $P \sim 0.0$) (which is similar to the Hellinger distance) (*30*). The last two metrics measure differences between probability distributions representing the topical content of the incoming tweet and the corpus of previous tweets to which users were exposed.

Although false rumors were measurably more novel than true rumors, users may not have perceived them as such. We therefore assessed users' perceptions of the information contained in true and false rumors by comparing the emotional content of replies to true and false rumors. We categorized the emotion in the replies by using the leading lexicon curated by the National Research Council Canada (NRC), which provides a comprehensive list of ~140,000 English words and their associations with eight emotions based on Plutchik's (*31*) work on basic emotion—anger, fear, anticipation, trust, surprise, sadness, joy, and disgust (*32*)—and a list of ~32,000 Twitter hashtags and their weighted associations with the same emotions (*33*). We removed stop words and URLs from the reply tweets and calculated the fraction of words in the tweets that related to each of the eight emotions, creating a vector of emotion weights for each reply that summed to one across the emotions. We found that false rumors inspired replies expressing greater surprise (K-S test = 0.205, $P \sim 0.0$), corroborating the novelty hypothesis, and greater disgust (K-S test = 0.102, $P \sim 0.0$), whereas the truth inspired replies that expressed greater sadness (K-S test = 0.037, $P \sim 0.0$), anticipation (K-S test = 0.038, $P \sim 0.0$), joy (K-S test = 0.061, $P \sim 0.0$), and trust (K-S test = 0.060, $P \sim 0.0$) (Fig. 4, D and F). The emotions expressed in reply to falsehoods may illuminate additional factors, beyond novelty, that inspire people to share false news. Although we cannot claim that novelty causes retweets or that novelty is the only reason why false news is retweeted more often, we do find that false news is more novel and that novel information is more likely to be retweeted.

Numerous diagnostic statistics and manipulation checks validated our results and confirmed their robustness. First, as there were multiple cascades for every true and false rumor, the variance of and error terms associated with cascades corresponding to the same rumor will be correlated. We therefore specified cluster-robust standard errors and calculated all variance statistics clustered at the rumor level. We tested the robustness of our findings to this specification by comparing analyses with and without clustered errors and found that, although clustering reduced the precision of our estimates as expected, the directions, magnitudes, and significance of our results did not change, and chi-square ($P \sim 0.0$) and deviance (d) goodness-of-fit tests ($d = 3.4649 \times 10^{-6}$, $P \sim 1.0$) indicate that the models are well specified (see supplementary materials for more detail).

Second, a selection bias may arise from the restriction of our sample to tweets fact checked by the six organizations we relied on. Fact checking may select certain types of rumors or draw additional attention to them. To validate the robustness of our analysis to this selection and the generalizability of our results to all true and false rumor cascades, we independently verified a second sample of rumor cascades that were not verified by any fact-checking organization. These rumors were fact checked by three undergraduate students at Massachusetts Institute of Technology (MIT) and Wellesley College. We trained the students to detect and investigate rumors with our automated rumor-detection algorithm running on 3 million English-language tweets from 2016 (*34*). The undergraduate annotators investigated the veracity of the detected rumors using simple search queries on the web. We asked them to label the rumors as true, false, or mixed on the basis of their research and to discard all rumors previously investigated by one of the fact-checking organizations. The annotators, who worked independently and were not aware of one another, agreed on the veracity of 90% of the 13,240 rumor cascades that they investigated and achieved a Fleiss' kappa of 0.88. When we compared the diffusion dynamics of the true and false rumors that the annotators agreed on, we found results nearly identical to those estimated with our main data set (see fig. S17). False rumors in the robustness data set had greater depth (K-S test = 0.139, $P \sim 0.0$), size (K-S test = 0.131, $P \sim 0.0$), maximum breadth (K-S test = 0.139, $P \sim 0.0$), structural virality (K-S test = 0.066, $P \sim 0.0$), and speed (fig. S17) and a greater number of unique users at each depth (fig. S17). When we broadened the analysis to include majority-rule labeling, rather than unanimity, we again found the same results (see supplementary materials for results using majority-rule labeling).

Third, although the differential diffusion of truth and falsity is interesting with or without robot, or bot, activity, one may worry that our conclusions about human judgment may be biased by the presence of bots in our analysis. We therefore used a sophisticated bot-detection algorithm (*35*) to identify and remove all bots before running the analysis. When we added bot traffic back into the analysis, we found that none of our main conclusions changed—false news still spread farther, faster, deeper, and more broadly than the truth in all categories of information. The results remained the same when we removed all tweet cascades started by bots, including human retweets of original bot tweets (see supplementary materials, section S8.3) and when we used a second, independent bot-detection algorithm (see supplementary materials, section S8.3.5) and varied the algorithm's sensitivity threshold to verify the robustness of our analysis (see supplementary materials, section S8.3.4). Although the inclusion of bots, as measured by the two state-of-the-art bot-detection algorithms we used in our analysis, accelerated the spread of both true and false news, it affected their spread roughly equally. This suggests that false news spreads farther, faster, deeper, and more broadly than the truth because humans, not robots, are more likely to spread it.

Finally, more research on the behavioral explanations of differences in the diffusion of true and false news is clearly warranted. In particular, more robust identification of the factors of human judgment that drive the spread of true and false news online requires more direct interaction with users through interviews, surveys, lab experiments, and even neuroimaging. We encourage these and other approaches to the investigation of the factors of human judgment that drive the spread of true and false news in future work.

False news can drive the misallocation of resources during terror attacks and natural disasters, the misalignment of business investments, and misinformed elections. Unfortunately, although the amount of false news online is clearly increasing (Fig. 1, C and E), the scientific understanding of how and why false news spreads is currently based on ad hoc rather than large-scale systematic analyses. Our analysis of all the verified true and false rumors that spread on Twitter confirms that false news spreads more pervasively than the truth online. It also overturns conventional wisdom about how false news spreads. Though one might expect network structure and individual characteristics of spreaders to favor and promote false news, the opposite is true. The greater likelihood of people to retweet falsity more than the truth is what drives the spread of false news, despite network and individual factors that favor the truth. Furthermore, although recent testimony before congressional committees on misinformation in the United States has focused on the role of bots in spreading false news (*36*), we conclude that human behavior contributes more to the differential spread of falsity and truth than automated robots do. This implies that misinformation-containment policies should also emphasize behavioral interventions, like labeling and incentives to dissuade the spread of misinformation, rather than focusing exclusively on curtailing bots. Understanding how false news spreads is the first step toward containing it. We hope our work inspires more large-scale research into the causes and consequences of the spread of false news as well as its potential cures.

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SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/359/6380/1146/suppl/DC1
Materials and Methods
Figs. S1 to S20
Tables S1 to S39
References (*37*–*75*)

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外 文 译 文

真假新闻的在线传播

Soroush Vosoughi, Deb Roy, Sinan Aral

麻省理工学院

第一章 概述

1.1 概述

决策、合作、通信和市场领域的基础理论全都将对真实或准确度的概念化作为几乎一切人类努力的核心。然而，不论是真实信息还是虚假信息都会于在线媒体上迅速传播。定义什么是真、什么是假成了一种常见的政治策略，而不是基于一些各方同意的事实的争论。我们的经济也难免遭受虚假信息传播的影响。虚假流言会影响股价和大规模投资的动向，例如，在一条声称巴拉克 · 奥巴马在爆炸中受伤的推文发布后，股市市值蒸发了 1300 亿美元。的确，从自然灾害到恐怖袭击，我们对一切事情的反应都受到了扰乱。

新的社交网络技术在使信息的传播速度变快和规模变大的同时，也便利了不实信息（即不准确或有误导性的信息）的传播。然而，尽管我们对信息和新闻的获取越来越多地收到这些新技术的引导，但我们仍然对他们在虚假信息传播上的作用知之甚少。尽管媒体对假新闻传播的轶事分析给予了相当多的关注，但仍然几乎没有针对不实信息扩散或其发布源头的大规模实证调查。目前，虚假信息传播的研究仅仅局限于小的、局部的样本的分析上，而这些分析忽略了两个最重要的科学问题：真实信息和虚假信息的传播有什么不同？哪些人类判断中的因素可以解释这些不同？

$$\max_{\boldsymbol{W}} tr(\boldsymbol{W}^{\mathrm{T}}\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}\boldsymbol{W}) \qquad \text{式（外 1-1）}$$

我只是为了把第二章挤到下一页而凑的字。我只是为了把第二章挤到下一页而凑的字。我只是为了把第二章挤到下一页而凑的字。

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第二章　我也不知道是什么

新的社交网络技术在使信息的传播速度变快和规模变大的同时，也便利了不实信息（即不准确或有误导性的信息）的传播。然而，尽管我们对信息和新闻的获取越来越多地收到这些新技术的引导，但我们仍然对他们在虚假信息传播上的作用知之甚少。尽管媒体对假新闻传播的轶事分析给予了相当多的关注，但仍然几乎没有针对不实信息扩散或其发布源头的大规模实证调查。目前，虚假信息传播的研究仅仅局限于小的、局部的样本的分析上，而这些分析忽略了两个最重要的科学问题：真实信息和虚假信息的传播有什么不同？哪些人类判断中的因素可以解释这些不同？

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北 京 邮 电 大 学

本科毕业设计（论文）开题报告

学院	计算机学院（国家示范性软件学院）		专业	计算机科学与技术	
学生姓名	宋健	学号	2022212702	班级	2022211305
指导教师姓名	金大海	所在单位	北京邮电大学	职称	副教授
设计（论文）题目	（中文）大模型驱动的代码质量多维度评估系统的设计与实现				
	（英文）Design and Implementation of Multidimensional Code Quality Evaluation System Driven by Large Language Models				

毕业设计（论文）开题报告内容：（主要包含选题的背景和意义；研究的基本内容和拟解决的主要问题；研究方法及措施；研究工作的步骤与进度；主要参考文献等项目。不少于 1500 字）

一、选题背景与意义

1.1 选题背景

随着大模型（Large Language Model, LLM）技术在软件工程领域的规模化应用，AI 辅助编程已成为工业级开发的主流模式。据 Shehab 等（2024）的研究数据显示，超过 40%的工业级代码开发依赖 AI 生成或辅助优化。然而，AI 生成代码在质量管控上仍面临严峻挑战，Sabra 等（2025）指出 AI 生成代码的漏洞率高达 37%，且在一致性、精简性等维度存在显著缺陷。

当前代码质量评估存在两类核心模式，但均存在明显短板：一是**传统自动评估模式**，以 SonarQube 为代表的工具依赖静态规则匹配，仅能覆盖语法错误、简单漏洞等基础指标，无法满足 ISO/IEC 25010:2023 国际标准中“功能适用性、性能效率、兼容性”等 9 大维度的完整评估需求，更缺乏对完整工程项目的跨文件依赖分析能力（香佳宏等，2025）；二是**纯 LLM 评估模式**，虽展现出语义理解优势，但存在关键缺陷：Pathak 等（2025）发现其缺乏专用评分标准（Rubric）导致评估结果主观性强；Wang 等（2025）的实证研究表明其对复杂质量维度的覆盖不全；且现有方案多为单体架构，未支持模型、评估标准、指标的灵活配置，也缺乏中间结果的追溯与可解释性（Tong 等，2024）。

在此背景下，构建一款完全遵循 ISO/IEC 25010:2023 标准、支持可配置化与微服务化、兼具高效性与可解释性的大模型驱动多维度代码质量评估系统，成为解决当前代码质量管控痛点的关键。

1.2 选题意义

（1）理论意义

- 突破传统“规则驱动”与“纯 LLM 驱动”的局限，建立“LLM+静态工具增强+ISO 全维度锚定”的混合评估理论体系，弥补现有研究在标准化、全维度覆盖方面的空白。
- 基于 Memory RAG 与知识图谱技术，解决 LLM 超长上下文处理难题，为大模型在大规模代码评估中的工程化应用提供理论支撑（Hou 等，2024）。

- 构建可解释、可追溯的自动化评估框架，通过 Agent 的 MCP（Multi-Capability Extension）能力扩展评估边界，丰富大模型在软件工程领域的应用方法论（Xia 等，2022）。

（2）实践意义

- 实现 **ISO 全维度覆盖**：严格遵循 ISO/IEC 25010:2023 标准，覆盖 9 大核心质量维度，解决传统工具维度覆盖不全的问题（ISO/IEC 25010:2023）。
- 提升评估效率与灵活性：通过可配置化设计（模型、标准、指标）适配不同场景，支持 20 万行以上代码的批量评估，大幅降低人工审查成本（Sun 等，2025）。
- 强化工程化实用性：微服务架构支持分布式部署与灵活扩展，生成包含“问题定位-推理依据-优化建议”的结构化报告，确保评估结果可追溯、可落地（Wong 等，2025）。

二、研究现状

现有代码质量评估研究围绕“传统自动评估-LLM 驱动评估”两个阶段演进，但均未形成兼顾 ISO 全维度、工程化、可解释性的解决方案，具体现状如下：

2.1 传统自动评估技术研究

- 以静态分析工具为核心，如 SonarQube 通过预设规则检测代码重复率、语法错误等指标，但仅能覆盖 ISO 标准中 30%的质量维度，缺乏对功能适用性、兼容性、交互能力等语义层面的评估（姜佳君等，2021）。
- 动态测试工具如 EvoSuite 通过生成测试用例验证正确性，但存在测试覆盖率不足（平均 60%以下）、无法适配复杂业务逻辑的问题（Fraser 等，2011）。
- 架构层面多为单体设计，不支持评估标准与指标的灵活配置，难以适配不同项目的个性化需求（Hou 等，2024）。

2.2 LLM 驱动评估研究

- 评估方法创新：Pathak 等（2025）提出“Rubric 驱动的 LLM 评估”方法，通过专用评分标准提升一致性；Tong 等（2024）开发的 CodeJudge 框架实现了多维度评估，但未覆盖 ISO 标准中的兼容性、交互能力等维度。
- 技术瓶颈：纯 LLM 评估存在三大缺陷：一是对 ISO 标准的适配不完整，仅聚焦正确性、可读性等少数维度；二是超长上下文处理能力不足，无法支持完整工程级评估；三是评估结果不可解释，缺乏中间推理过程与证据链（香佳宏等，2025）。

2.3 标准化与工程化研究

ISO/IEC 25010:2023 标准明确了 9 大核心质量维度及细分指标，但现有评估工具均未完全覆盖，且缺乏将标准指标转化为可计算、自动化评估逻辑的成熟方案（ISO/IEC 25010:2023）；工程化方面，现有系统多未采用微服务架构，扩展性差，且未实现模型、指标、评估标准的可配置化（Sun 等，2025）。

三、研究的基本内容和拟解决的主要问题

3.1 研究基本内容

（1）大模型选型与增强方案设计

- 集成 Memory RAG 技术，将代码依赖关系、中间评估结果存入知识库，解决超长上下文处理问题；引入轻量级代码结构图谱（Code Structure Graph），构建“文件-类-函数”关联网络，提升跨文件依赖评估的准确性（Zeng 等，2022）。
- 为 Agent 配置 MCP 能力，集成静态分析工具（Tree-sitter）、性能测试工具（Locust）、漏洞检测工具（CodeQL），扩展评估工具链覆盖范围。

（2）ISO 全维度评估指标体系构建（基于 ISO/IEC 25010:2023）

严格遵循 ISO 标准，构建 9 大核心维度+细分指标的评估体系，每个指标明确自动化计算思路：

1. 功能适用性
    - 完整性：满足的功能点数量/项目特性列表（README）功能点数量（通过 LLM 提取 README 特性与代码实现匹配）。
    - 正确性：自动生成单测的通过率（基于 Pynguin/EvoSuite 生成测试用例，通过率≥配置阈值（默认 90%）为合格）（Fraser 等，2011）。
    - 适当性：代码功能与预期场景的匹配度（LLM 计算代码注释/文档与代码语义相似度，≥配置阈值（默认 85%）为合格）。
2. 性能效率
    - 时间行为：接口平均响应时间/基准版本响应时间（通过 Locust 模拟并发请求测试，比值≤配置阈值（默认 1.2）为合格）。
    - 资源利用率：内存峰值占用/基准版本占用（通过 Docker 监控，无显著异常增长为合格）。
    - 容量：系统稳定支持的最大并发用户数/设计阈值（通过压力测试，≥配置阈值（默认 90%）为合格）。
3. 兼容性
    - 共存性：代码在多环境（Windows/Linux/macOS）的编译通过率（≥配置阈值（默认 95%）为合格）。
    - 互操作性：与依赖组件的接口调用成功率（通过 Mock 测试，≥配置阈值（默认 99%）为合格）。
4. 交互能力
    - 可学习性：API 文档完整性+命名规范符合度（文档覆盖率≥配置阈值（默认 90%）且命名规范符合率≥配置阈值（默认 85%）为合格）。
    - 可操作性：接口调用复杂度（平均调用步骤≤配置阈值（默认 3 步）为合格）。
    - 包容性：代码对特殊输入（空值、异常格式）的处理能力（无崩溃且返回合理提示率≥配置阈值（默认 90%）为合格）。
5. 可靠性
    - 无故障性：连续运行 24 小时的故障次数（≤配置阈值（默认 1 次）为合格）。
    - 可用性：系统正常运行时间占比（≥配置阈值（默认 99.9%）为合格）。
    - 容错性：异常场景下的系统恢复时间（≤配置阈值（默认 5 秒）为合格）。
6. 安全性

- 保密性：敏感数据（密码、密钥）硬编码检测通过率（通过 CodeQL 检测，通过率≥配置阈值（默认 100%）为合格）。
- 完整性：关键数据操作的校验逻辑覆盖率（≥配置阈值（默认 90%）为合格）。
- 抗性：常见漏洞模式（CWE Top 25）的扫描通过率（通过 CodeQL 静态扫描，无高危漏洞为合格）。

7. 可维护性
   - 模块性：无循环依赖的模块占比（≥配置阈值（默认 90%）为合格，通过依赖分析工具检测）。
   - 可分析性：代码注释覆盖率（≥配置阈值（默认 30%）为合格）。
   - 可测试性：单元测试覆盖率（≥配置阈值（默认 70%）为合格）。
8. 灵活性
   - 适应性：代码适配不同业务场景的修改量（平均修改行数≤配置阈值（默认 10 行）为合格）。
   - 可扩展性：新增功能时的代码复用率（≥配置阈值（默认 60%）为合格）。
   - 可替换性：核心模块的接口标准化程度（≥配置阈值（默认 90%）为合格）。
9. 安全（**Safety**）
   - 风险识别：高风险代码片段占比（≤配置阈值（默认 5%）为合格，通过 LLM 结合安全规则检测）。
   - 故障安全：故障时数据丢失率（≤配置阈值（默认 0）为合格）。
   - 安全集成：与安全组件的兼容适配度（无冲突且功能正常为合格）。

（3）可配置化微服务架构设计

- 微服务拆分（基于 Python FastAPI）：拆分为模型管理服务、指标配置服务、代码解析服务、评估执行服务、报告生成服务、结果追溯服务、数据存储服务。
- 可配置化实现：支持模型类型（GPT-4/CodeLlama 等）、评估标准（ISO 标准/企业自定义）、指标权重的动态配置（通过配置中心实现）（Sun 等，2025）。
- 中间结果追溯：记录每个评估维度的推理过程、数据来源、工具输出日志，支持问题溯源与全链路查询。

（4）评估流程优化与自动化验证

- 专用提示词工程：为 9 大维度及细分指标分别设计 Rubric 驱动的提示词，强制模型输出“问题定位-推理依据-评分结果”结构化数据（Pathak 等，2025）。
- 评估流程设计：代码解析→上下文提取→多工具协同评估→LLM 整合结果→结构化报告生成。
- 大规模验证：构建包含 50 个开源项目（涵盖 Java、Python、C++）的测试数据集（累计代码量≥20 万行），与 SonarQube 等主流工具对比，验证系统准确率与维度覆盖度。

3.2 拟解决的主要问题

1. 传统评估工具 ISO 维度覆盖不全、纯 LLM 评估主观偏差大的问题，通过 ISO 全维度锚定+混合评估架构实现精准评估。
2. 超长上下文处理与跨文件依赖分析难题，通过 Memory RAG+知识图谱技术突破限制。
3. 评估结果不可解释、不可追溯的问题，通过结构化报告与中间过程日志实现全链路可查。
4. 系统灵活性与扩展性不足的问题，通过微服务化与可配置化设计适配不同应用场景。

四、研究方法及措施

4.1 阶段一：前期准备

1. 文献与标准调研：系统研读 ISO/IEC 25010:2023 标准、10 篇以上核心参考文献，梳理评估指标与技术路线。
2. 数据集构建：收集 50 个开源项目（Java/Python/C++各不少于 15 个），累计代码量≥20 万行，整理项目 README/文档、引入 Defects4J/QuixBugs 等包含已知缺陷的数据集作为评估基准。
3. 工具选型与环境搭建：确定开发工具（Vue3、FastAPI、Tree-sitter、Locust）、模型接口（OpenAI API/CodeLlama 开源部署）、数据库（PostgreSQL、Redis），搭建本地开发与测试环境。

4.2 阶段二：核心技术预研

1. 大模型选型测试：设计测试用例（涵盖 9 大维度典型场景），对比 CodeLlama、GPT-4 Code、Qwen-Coder 的评估准确率与推理速度，筛选最优模型。
2. Memory RAG 与知识图谱实现：
    - 基于 LangChain 构建 Memory RAG 模块，实现代码片段、依赖关系的存储与快速查询。
    - 使用 Neo4j 构建代码结构图谱，定义“文件-类-函数”关联关系，支持跨文件依赖分析。
3. Agent MCP 能力集成：开发工具调用接口，实现 LLM 对 Tree-sitter（代码解析）、Locust（性能测试）、CodeQL（漏洞检测）的自动调用与结果整合。

4.3 阶段三：系统设计

1. 微服务架构设计：绘制系统架构图、模块流程图、接口文档，明确各服务职责与交互逻辑。
2. 可配置化方案设计：设计模型配置表、指标权重配置表、评估标准配置表，实现动态调整功能。
3. 评估流程与提示词设计：
    - 设计“代码解析→多工具评估→LLM 整合→报告生成”的自动化流程。
    - 为 9 大维度及细分指标设计专用提示词，确保输出结构化数据。
4. 数据库设计：设计评估任务表、中间结果表、报告表、日志表，支持数据追溯与查询。

4.4 阶段四：系统开发

1. 后端服务开发：
    - 模型管理服务：实现模型注册、调用、负载均衡功能。
    - 指标配置服务：支持 ISO 指标的启用/禁用、权重调整。
    - 代码解析服务：集成 Tree-sitter 与知识图谱，实现代码结构与依赖关系提取。
    - 评估执行服务：集成 Agent MCP 能力，实现多工具协同评估与 LLM 结果整合。
    - 报告生成服务：生成可视化报告（雷达图、缺陷分布热力图）与结构化文档。
    - 结果追溯服务：记录评估全流程日志，支持中间结果查询。
2. 前端开发：实现项目上传、配置管理、评估任务提交、报告查看、结果追溯等功能。

3. 集成测试：解决服务间接口调用问题，确保系统流程顺畅运行。

4.5 阶段五：系统测试与优化

1. 功能测试：针对 9 大维度设计测试用例，验证每个指标的计算准确性与评估流程完整性。
2. 性能测试：通过 20 万行代码批量评估，测试系统响应时间（≤30 分钟）、并发处理能力（支持 5 个并行任务）。
3. 准确率优化：与 SonarQube 等工具对比评估结果，调整提示词与指标计算逻辑，将评估准确率提升至 90%以上。
4. 兼容性测试：验证系统在不同操作系统（Windows/Linux/macOS）、浏览器（Chrome/Firefox/Edge）的运行稳定性。

4.6 阶段六：系统验证与论文撰写

1. 大规模数据集验证：使用构建的 50 个开源项目数据集进行评估，统计系统的维度覆盖度、准确率、效率指标。
2. 论文撰写：整理实验数据、系统设计文档、测试报告，撰写毕业论文，完成摘要、引言、文献综述、系统设计、实验验证等章节。
3. 答辩准备：制作答辩 PPT，梳理系统创新点、设计方案与实验成果，准备演示流程。

五、研究工作的步骤与进度

周次	核心任务	具体产出
第 1-2 周	文献与 ISO 标准调研	文献综述报告、ISO 指标拆解文档
第 3-4 周	数据集构建与环境搭建	50 个开源项目数据集、开发环境部署完成
第 5-6 周	大模型选型测试	模型对比测试报告、最优模型确定
第 7-8 周	Memory RAG 与知识图谱实现	RAG 模块代码、知识图谱构建脚本
第 9-10 周	微服务架构与可配置化设计	架构图、接口文档、配置表设计
第 11-12 周	评估流程与提示词设计	流程设计文档、9 大维度专用提示词库
第 13-14 周	模型管理服务+指标配置服务开发	服务代码、单元测试用例
第 15-16 周	代码解析服务+知识图谱集成	服务代码、代码依赖分析功能实现
第 17-18 周	评估执行服务+Agent MCP 集成	服务代码、工具自动调用功能实现
第 19-20 周	报告生成服务+结果追溯服务开发	服务代码、报告生成与日志查询功能
第 21-22 周	前端开发	前端页面、前后端联调完成

第 23-24 周	系统集成测试	集成测试报告、Bug 修复完成
第 25-26 周	功能测试与准确率优化	功能测试报告、准确率优化记录
第 27-28 周	性能测试与兼容性测试	性能测试报告、兼容性测试报告
第 29-30 周	大规模数据集验证	验证报告、对比分析文档
第 31-32 周	论文撰写与答辩准备	毕业论文初稿、答辩 PPT

六、预期成果

1. **原型系统一套**：大模型驱动的代码质量多维度评估系统（微服务架构），支持 ISO/IEC 25010:2023 全维度评估，可配置化模型、指标、评估标准，评估准确率≥90%。
2. **毕业论文一篇**：字数约 2.5 万字，系统阐述 ISO 指标体系构建、核心技术实现、系统设计与实验验证结果。
3. **实验验证报告一份**：包含 50 个开源项目的测试数据、与主流工具的对比结果、系统性能指标（响应时间、并发能力）。
4. **专用资源库**：ISO 指标计算手册、专用提示词库、测试数据集（含评估基准）、系统源码与部署文档。

七、主要参考文献

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日期	2025 年 12 月 26 日	签字	

注：可根据开题报告的长度加页

北 京 邮 电 大 学

本科毕业设计（论文）中期进展情况检查表

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教师指导本科毕业设计（论文）记录表

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注：指导教师应从选题后开始指导并记录，直至学生提交定稿论文。每 2 周指导内容记录在一个表格中，双面打印。